Assignment 5 - Protocol Analysis With Wireshark

Objective

In this lab, your objective of this lab to gain familiarity with the TCP/IP Layer 2, 3, and 4 headers in a live environment using Wireshark.

Learning Outcomes

At the end of this section, you will be able to:

1. Install, configure, and deploy Wireshark Protocol Analyzer
2. View and analyze TCP/IP Layer 2,3, and 4 headers using Wireshark in a live environment

Part 1 – Install and Configure Wireshark

1. On your computer, use the provided installer to install Wireshark Protocol Analyzer and verify its functionality.

2. Before proceeding, review and familiarize yourself with the available Wireshark documentation:

   a. https://www.wireshark.org/docs/wsug_html_chunked/

   b. https://youtu.be/qiFcd7WA6mk?si=EaSDsuqKL6ReUx9l

   c. https://www.wireshark.org/docs/

   d. https://gitlab.com/wireshark/wireshark/-/wikis/home

   e. How to Filter HTTP Traffic in Wireshark | NetworkProGuide

Part 2 – Analyze Live Traffic with Wireshark

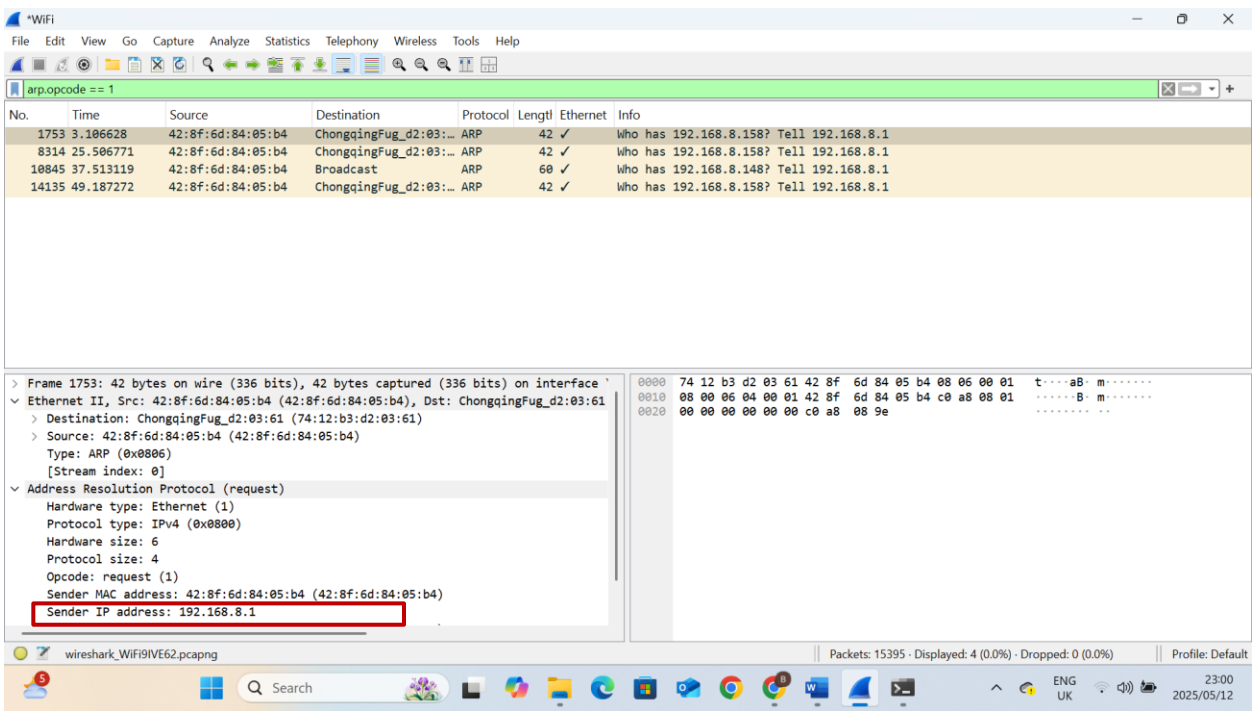
1. Run a Wireshark capture on the Ethernet adapter for 1-2 minutes, and then stop your capture.

2. Use the available Wireshark filters to isolate and view the following packets. (Note: You may have to restart your capture and specifically generate specific traffic to capture your desired packets).

3. All packets should be verified with screenshots that are pasted into your Word document below. Note: You might be able to show more than one answer with one screenshot. Please be sure that your screenshot makes the packet clearly identifiable.

Part 3 – Deliverables

1. Isolate and view an **ARP Request** packet.



a. What are the source and destination Layer 2 addresses for this packet?

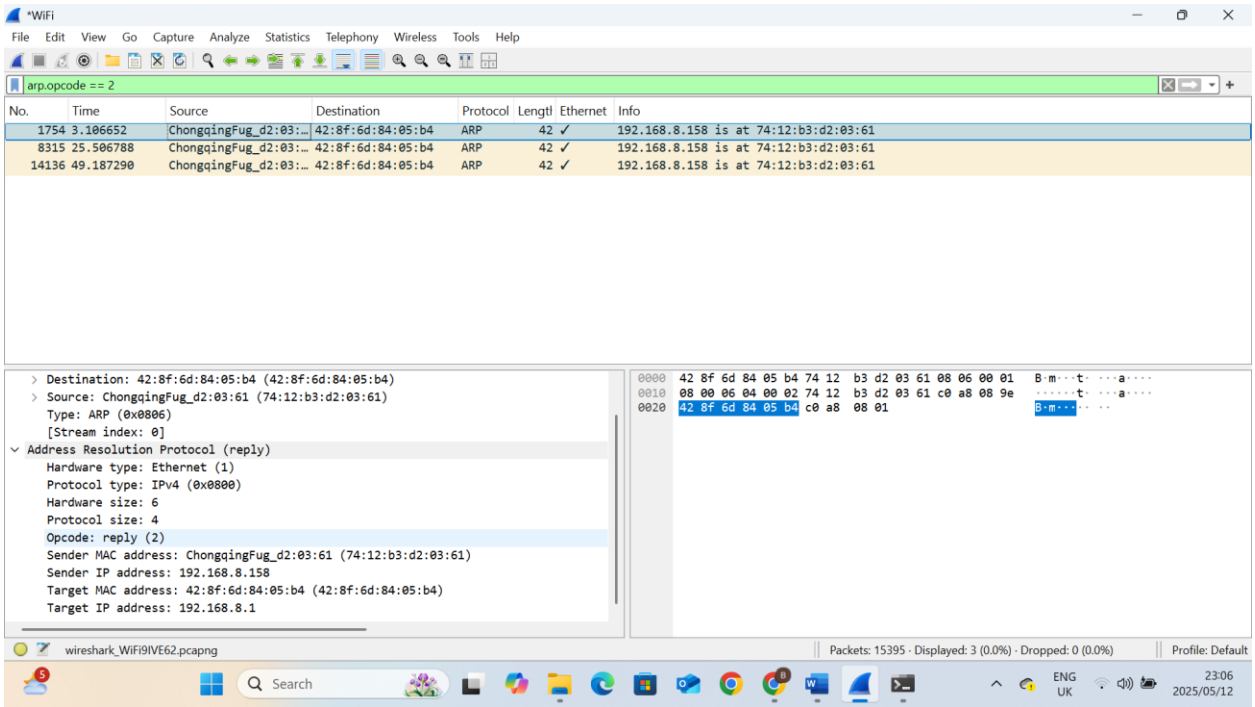
Source Address: 42:8f:6b:84:05:b4

Destination Address: 74:12:b3:d2:03:61

b. What is the source IP of this packet?

192.168.8.1

2. Isolate and view the **ARP Reply** packet to the request that you isolated in question 1.



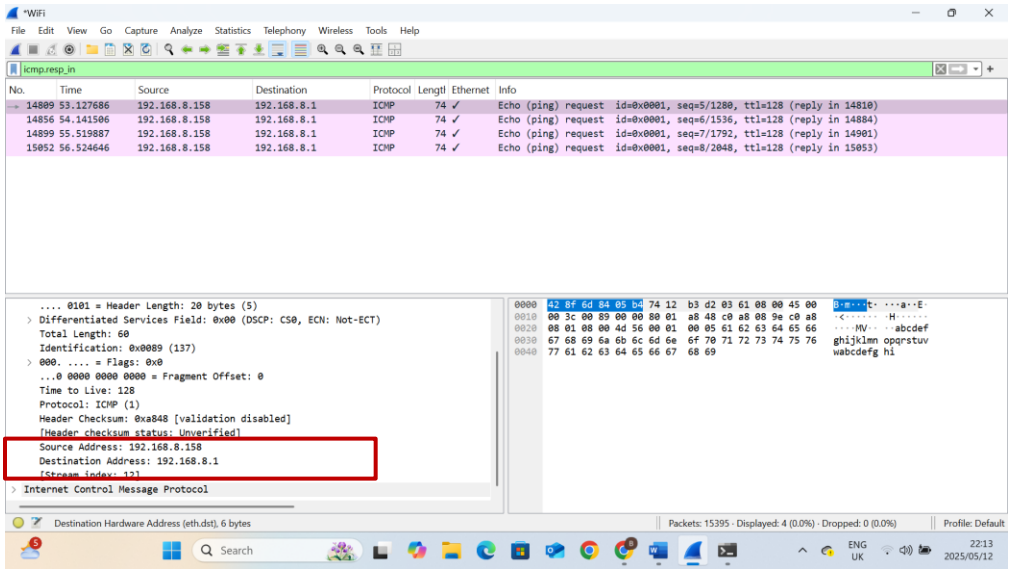
   a. How can you verify that this reply is in response to the original request?

      Target IP & MAC Address in the reply should match the original Sender IP and MAC in the request.

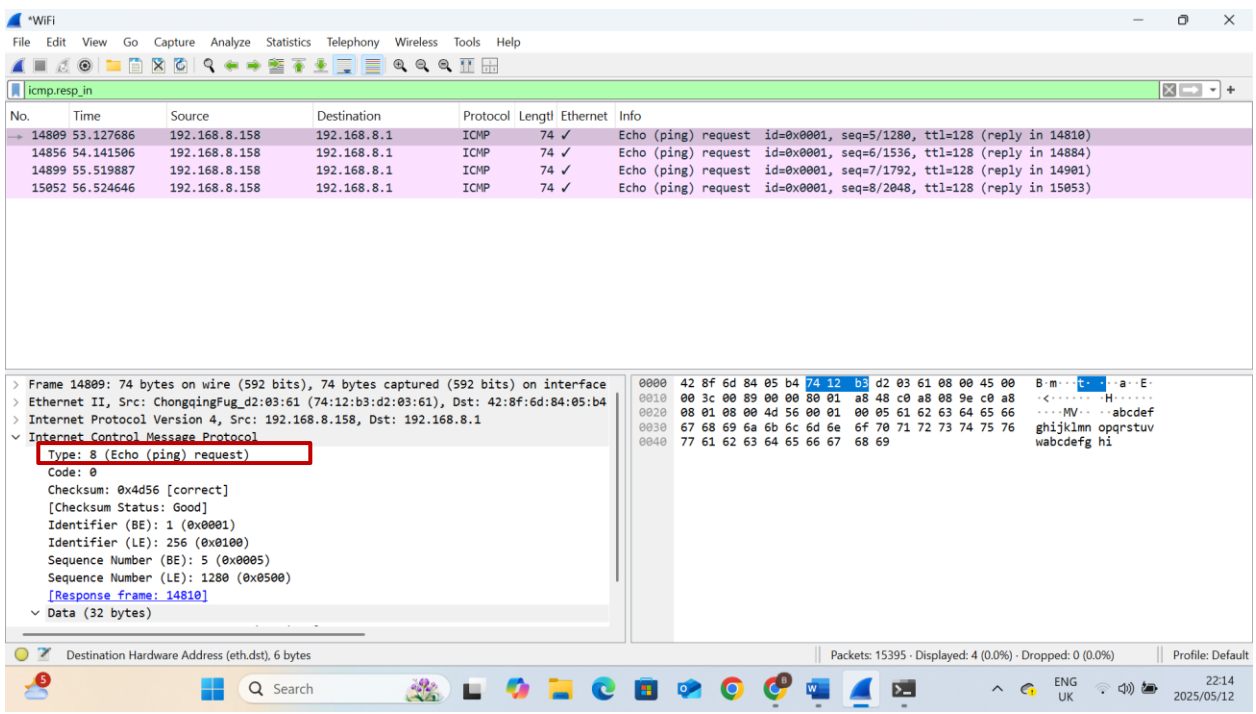
3. Isolate and view a **PING Request** packet.

   a. Verify in the packet where you would find the source and destination Layer 3 addresses.

   Expand the Internet Protocol Version 4 section in the packet details.



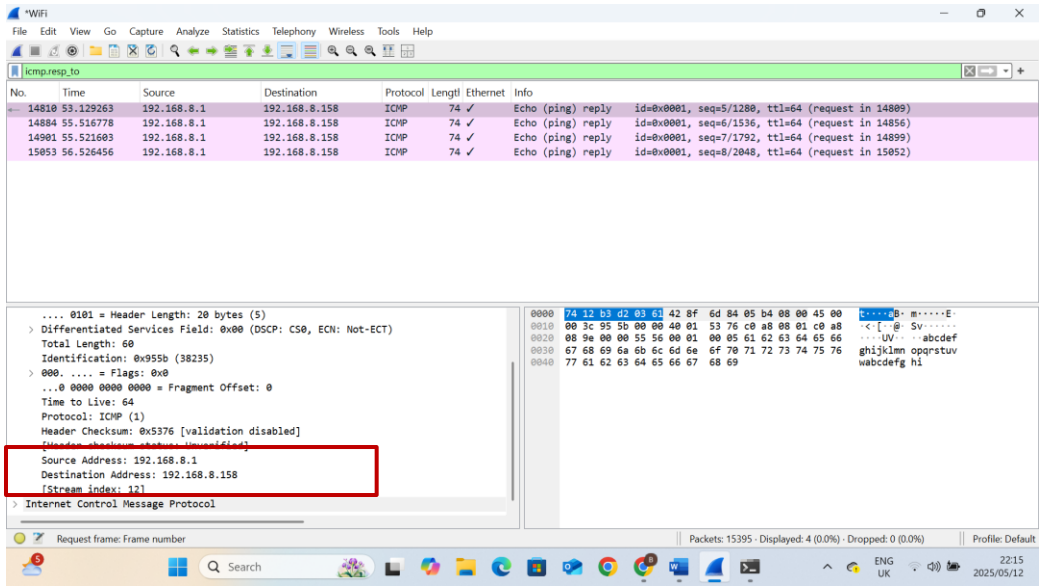
b. What is the ICMP "Type" of this packet? Type: 8 (Echo (ping) request)



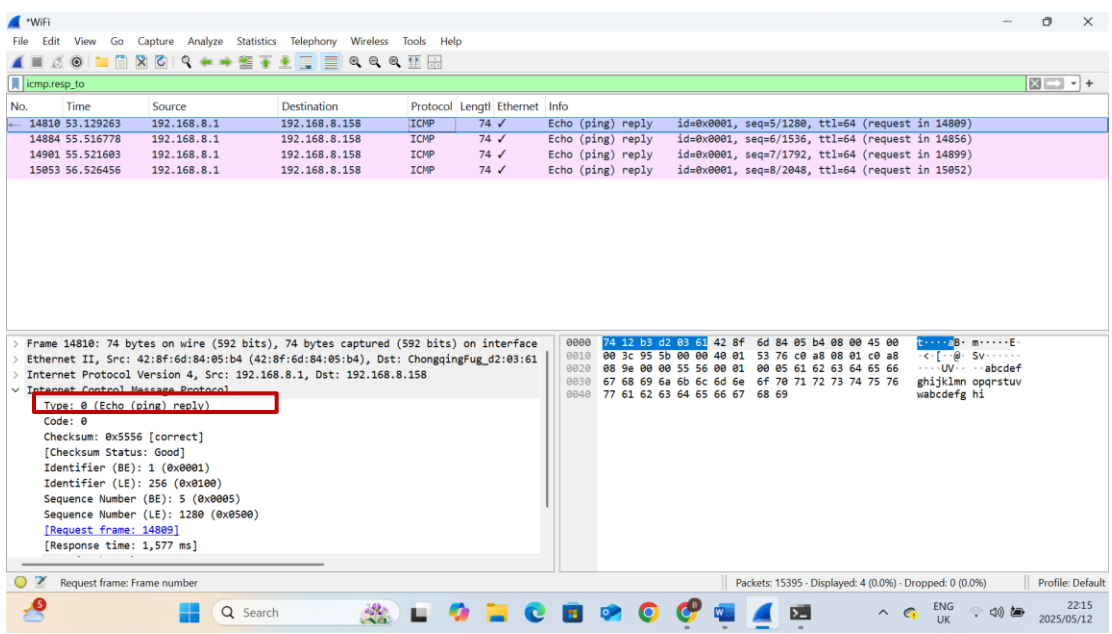
4. Isolate and view a **PING Reply** packet.

a. Verify in the packet where you would find the source and destination Layer 3 addresses.

Expand the Internet Protocol Version section in the packet details.



b. What is the ICMP "Type" of this packet? Type: 0 (Echo (ping) reply)

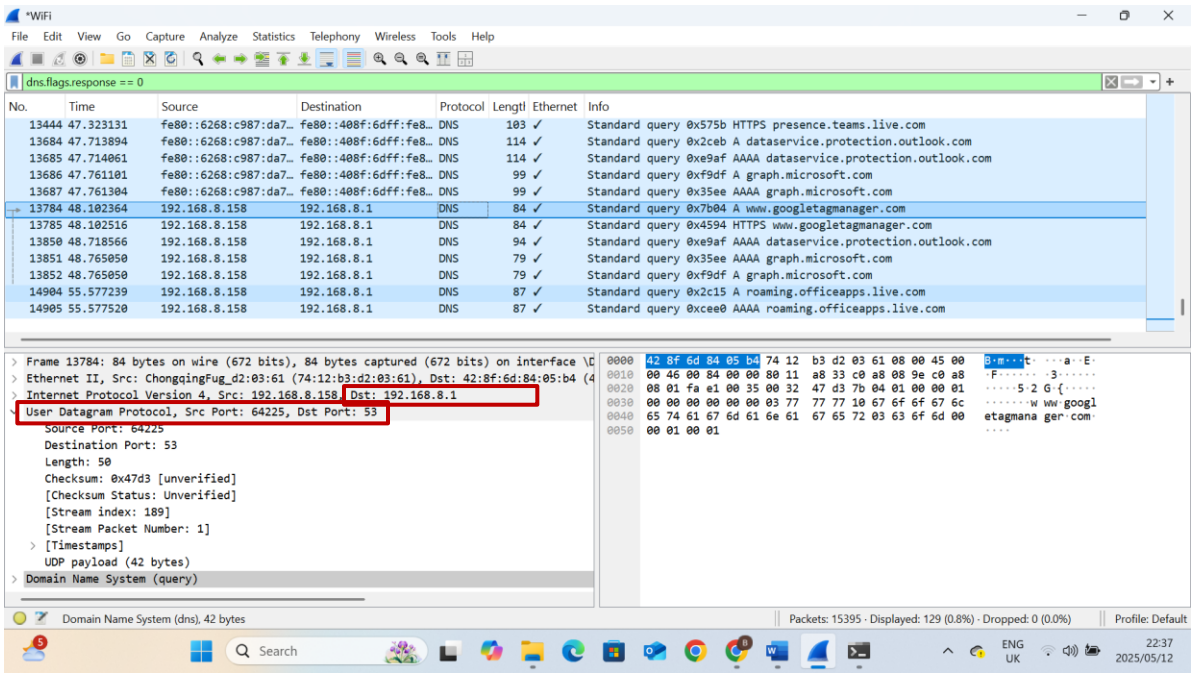


c. Use the Internet to determine what other IMCP "types" there are, and list them below.

Type 0 — Echo Reply
Type 1 — Unassigned
Type 2 — Unassigned
Type 3 — Destination Unreachable
Type 4 — Source Quench (Deprecated)
Type 5 — Redirect
Type 6 — Alternate Host Address (Deprecated)
Type 7 — Unassigned
Type 8 — Echo
Type 9 — Router Advertisement
Type 10 — Router Selection
Type 11 — Time Exceeded
Type 12 — Parameter Problem
Type 13 — Timestamp
Type 14 — Timestamp Reply
Type 15 — Information Request (Deprecated)
Type 16 — Information Reply (Deprecated)
Type 17 — Address Mask Request (Deprecated)
Type 18 — Address Mask Reply (Deprecated)
Type 19 — Reserved (for Security)
Types 20-29 — Reserved (for Robustness Experiment)
Type 30 — Traceroute (Deprecated)
Type 31 — Datagram Conversion Error (Deprecated)
Type 32 — Mobile Host Redirect (Deprecated)
Type 33 — IPv6 Where-Are-You (Deprecated)
Type 34 — IPv6 I-Am-Here (Deprecated)
Type 35 — Mobile Registration Request (Deprecated)
Type 36 — Mobile Registration Reply (Deprecated)
Type 37 — Domain Name Request (Deprecated)
Type 38 — Domain Name Reply (Deprecated)
Type 39 — SKIP (Deprecated)
Type 40 — Photuris
Type 41 — ICMP messages utilized by experimental mobility protocols such as Seamoby
Type 42 — Extended Echo Request
Type 43 — Extended Echo Reply

Types 44-252 — Unassigned
Type 253 — RFC3692-style Experiment 1
Type 254 — RFC3692-style Experiment 2

5. Isolate and view a **DNS Query** packet.



a. Is this packet UDP or TCP?

UDP

b. What are the source and destination ports being used by this packet?

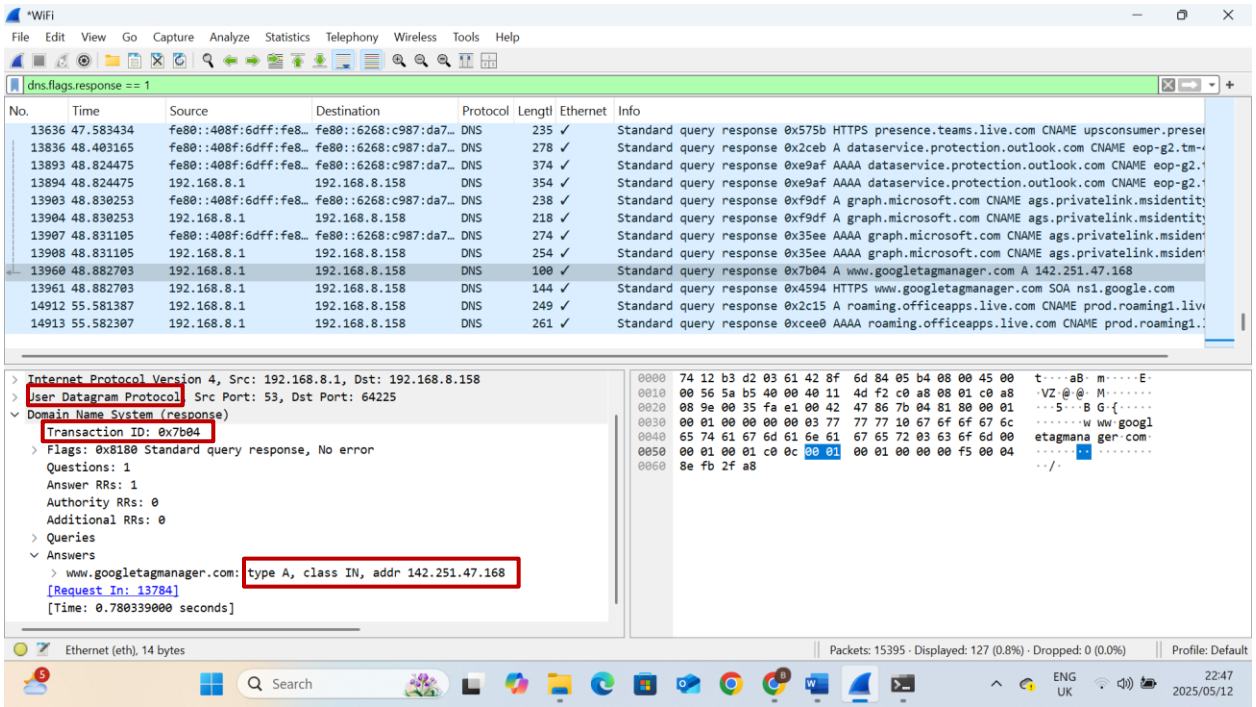
Source ports: 64 225

Destination ports: 53

c. What is the IP address of the destination server?

IP address: 192.168.8.1

6. Isolate and view a **DNS Response** to the DNS query above.



a. Is this packet UDP or TCP?

UDP

b. How can you verify that this reply is in response to the original request?

Transaction ID: 0x7b04

c. What type of DNS record was returned by the destination server?

Type A

d. What is the IP address of the server who is listed in the record that is included in the DNS response? Note: depending on the capture, this may or may not be the same as the destination IP address in the packet.

Address 142.251.47.168

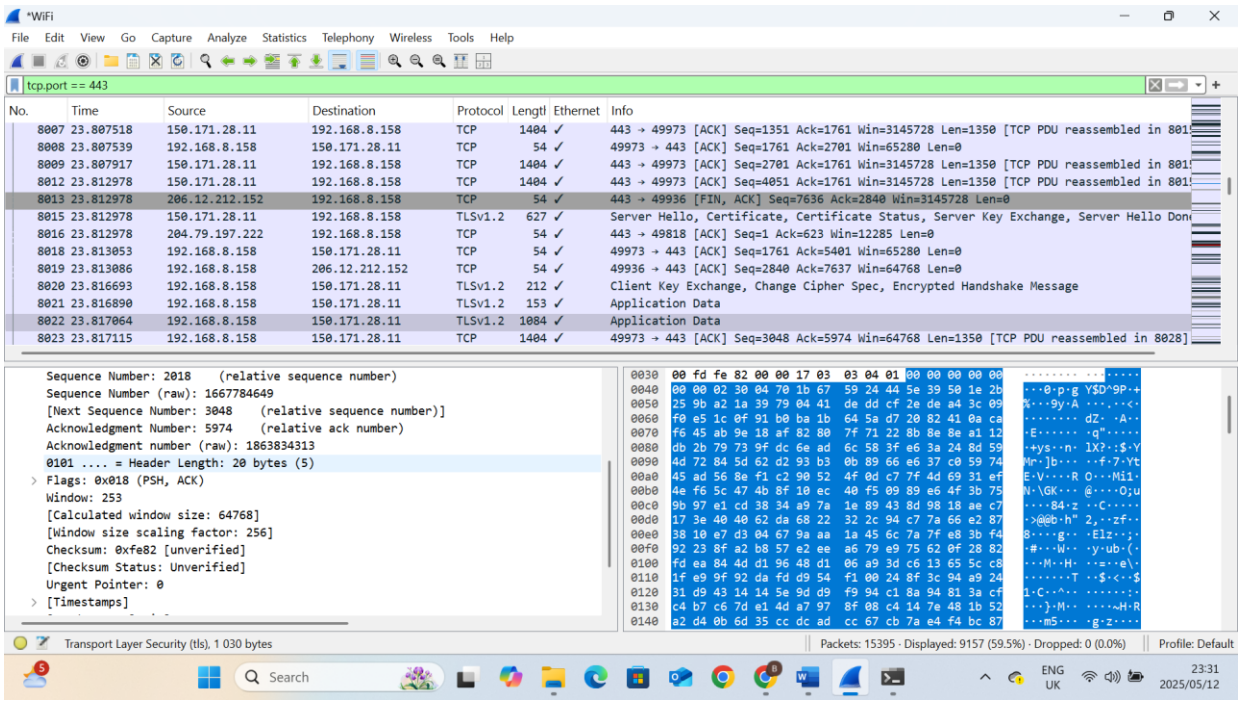
7. Filter your capture to review all traffic to or from port 443.

a. Are these packets UDP or TCP? What type of data are they requesting and retrieving?

TCP

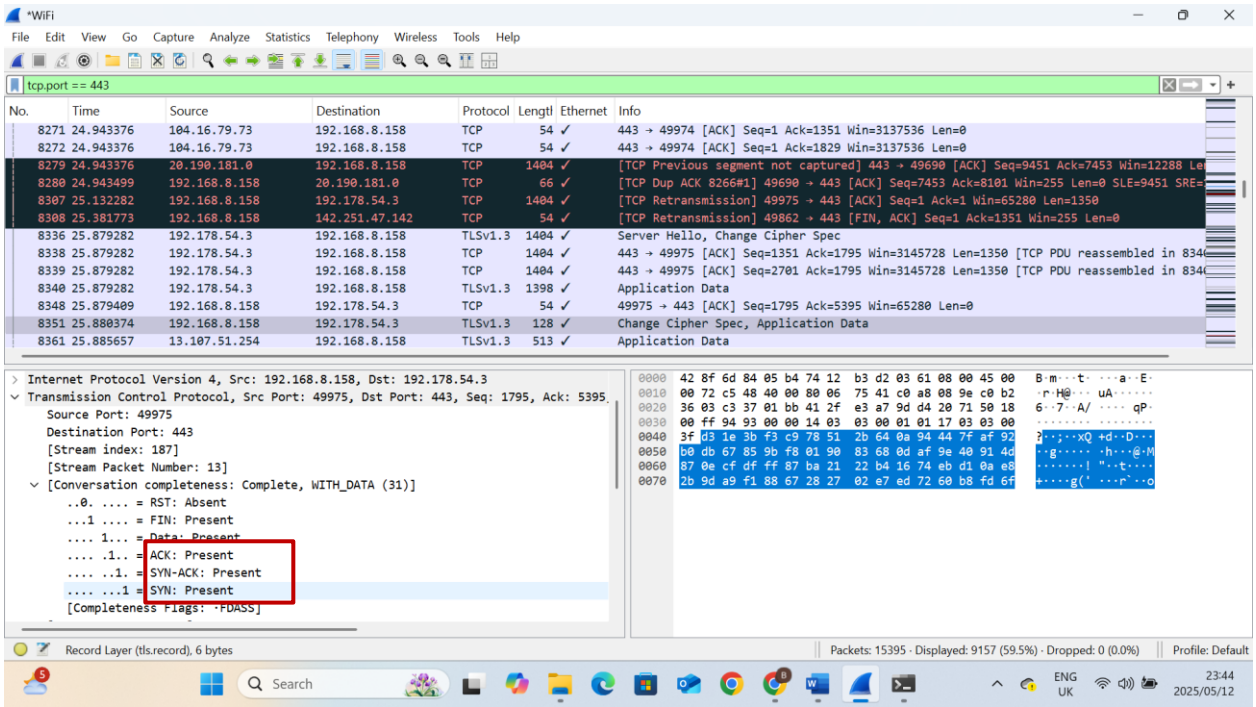
b. Open and review a packet that contains **Application** data. Where in the packet can you confirm that the payload is encrypted?

you cannot view the actual contents



c. Many of these packets are TCP control packets that establish and maintain the reliable TCP connection. By looking at the packets that contain flags (as well as using your Netacad text or the Internet as an additional source), briefly describe the process called the three-way TCP handshake. Support your answer with screenshots of packets that contain the appropriate flags.

1. SYN
2. SYN-ACK
3. ACK



Lab Submission

Using the lab submission link in Moodle, upload a copy of this Word document with the requested screenshots.